

# Report Summary

ML Tactical Intelligence Report

Generated: 2026-05-21 04:20:43 UTC

Outputs include quality report, metrics, cluster summary, and charts.

# Quality Report

```
{
  "total_rows": 2720,
  "total_columns": 27,
  "columns": [
    "key_id",
    "goal_id",
    "tournament_id",
    "tournament_name",
    "match_id",
    "match_name",
    "match_date",
    "stage_name",
    "group_name",
    "team_id",
    "team_name",
    "team_code",
    "home_team",
    "away_team",
    "player_id",
    "family_name",
    "given_name",
    "shirt_number",
    "player_team_id",
    "player_team_name",
    "player_team_code",
    "minute_label",
    "minute_regulation",
    "minute_stoppage",
    "match_period",
    "own_goal",
    "penalty"
  ],
  "dtypes": {
    "key_id": "int64",
    "goal_id": "object",
    "tournament_id": "object",
    "tournament_name": "object",
    "match_id": "object",
    "match_name": "object",
    "match_date": "object",
    "stage_name": "object",
    "group_name": "object",
    "team_id": "object",
    "team_name": "object",
    "team_code": "object",
    "home_team": "int64",
    "away_team": "int64",
    "player_id": "object",
    "family_name": "object",
    "given_name": "object",
    "shirt_number": "int64",
    "player_team_id": "object",
    "player_team_name": "object",
    "player_team_code": "object",
    "minute_label": "object",
    "minute_regulation": "int64",
    "minute_stoppage": "int64",
    "match_period": "object",
    "own_goal": "int64",
    "penalty": "int64"
  },
  "null_counts": {
    "key_id": 0,
    "goal_id": 0,
    "tournament_id": 0,
    "tournament_name": 0,
    "match_id": 0,
    "match_name": 0,
    "match_date": 0,
    "stage_name": 0,
    "group_name": 0,
    "team_id": 0,
    "team_name": 0,
    "team_code": 0,
    "home_team": 0,
```

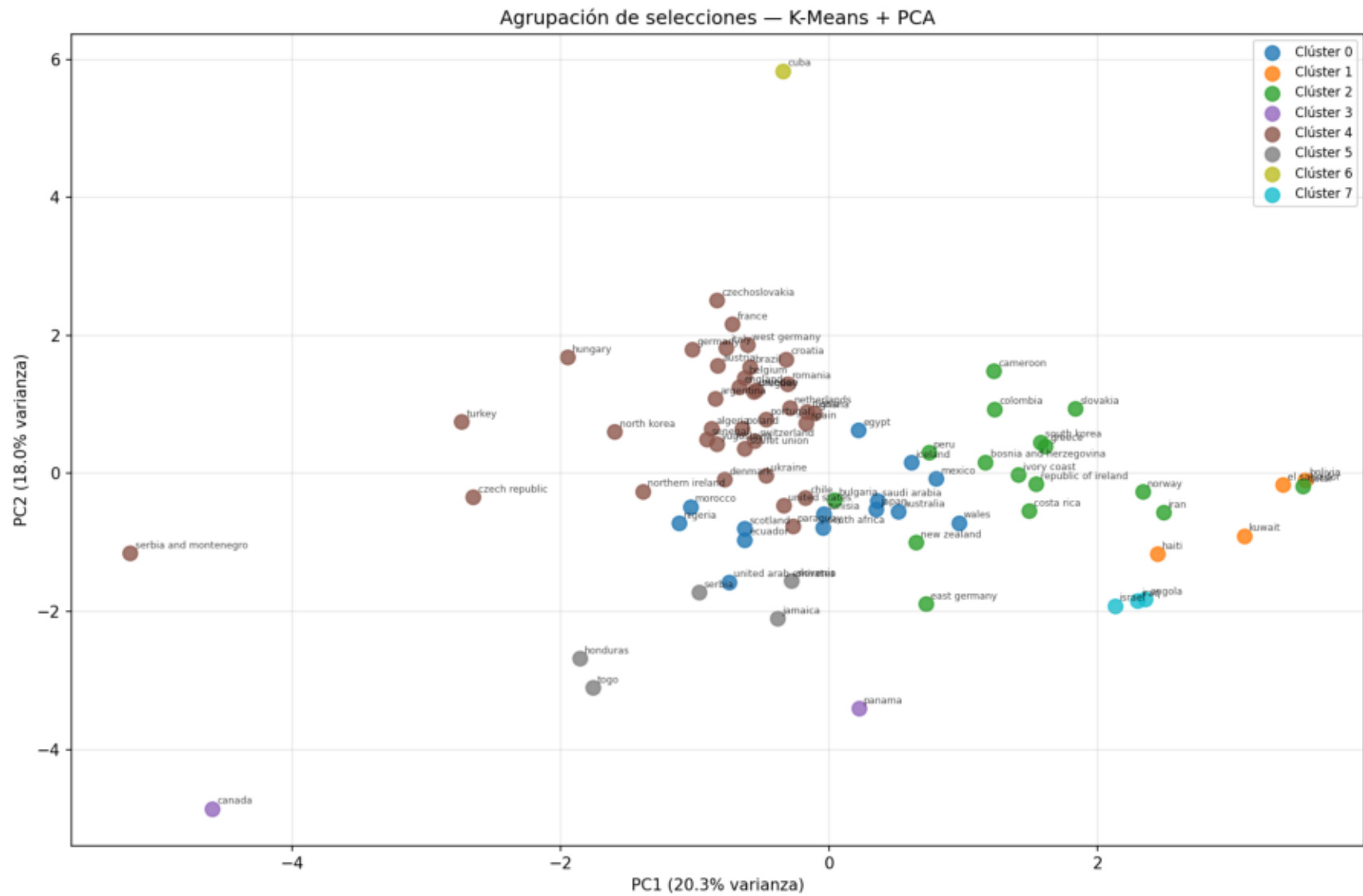
# Metrics Report

```
{
  "clustering": {
    "k": 8,
    "silhouette": 0.21894739937385943,
    "davies_bouldin": 1.1144772714909963
  },
  "random_forest": {
    "cross_validation": {
      "cv_accuracy_mean": 0.7309,
      "cv_accuracy_std": 0.0079,
      "cv_f1_mean": 0.7307,
      "cv_f1_std": 0.008,
      "cv_auc_mean": 0.8067,
      "cv_auc_std": 0.0097
    },
    "validation_set": {
      "accuracy": 0.6483,
      "precision_macro": 0.555,
      "precision_weighted": 0.6775,
      "recall_macro": 0.5635,
      "recall_weighted": 0.6483,
      "f1_macro": 0.5563,
      "f1_weighted": 0.6608,
      "auc_roc": 0.6209
    },
    "test_set": {
      "accuracy": 0.6388,
      "precision_macro": 0.5519,
      "precision_weighted": 0.6844,
      "recall_macro": 0.5636,
      "recall_weighted": 0.6388,
      "f1_macro": 0.5513,
      "f1_weighted": 0.657,
      "auc_roc": 0.6295
    }
  }
}
```

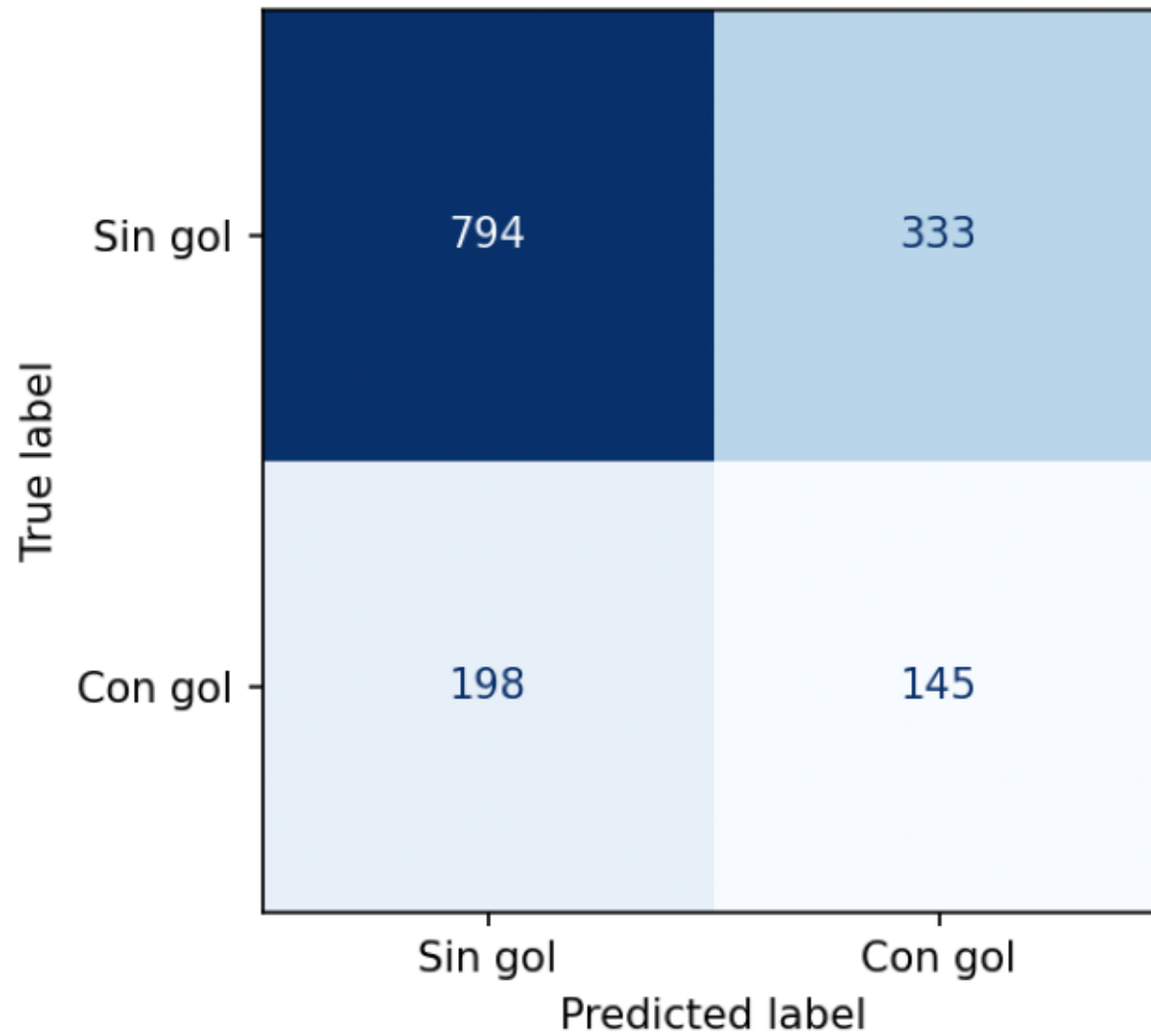
## Cluster Summary

|            | total_goals  | matches_played | pct_0_15            | pct_16_30           | pct_31_45   | pct_46_60 | pct_61_75 | pct_76_90 |
|------------|--------------|----------------|---------------------|---------------------|-------------|-----------|-----------|-----------|
| pct_90plus | penalty_rate | own_goal_rate  | avg_goals_per_match | knockout_goal_ratio | mean_minute |           |           |           |
| cluster    |              |                |                     |                     |             |           |           |           |
| 0          | 17.0714      | 11.0714        | 8.5835              | 23.6908             | 15.0204     | 19.2791   | 11.5667   | 21.8595   |
| 0.0000     | 0.1732       | 0.0102         |                     | 1.4410              |             | 0.0690    | 47.6872   |           |
| 1          | 1.5000       | 1.5000         | 0.0000              | 0.0000              | 0.0000      | 25.0000   | 75.0000   | 0.0000    |
| 0.0000     | 0.0000       | 0.0000         |                     | 1.0000              |             | 0.0000    | 62.8750   |           |
| 2          | 14.0625      | 9.1250         | 7.6294              | 6.0028              | 9.9537      | 15.6468   | 21.7727   | 38.0707   |
| 0.9238     | 0.0854       | 0.0319         |                     | 1.5177              |             | 0.0939    | 62.7856   |           |
| 3          | 2.0000       | 2.0000         | 25.0000             | 0.0000              | 50.0000     | 0.0000    | 0.0000    | 25.0000   |
| 0.0000     | 0.0000       | 0.5000         |                     | 1.0000              |             | 0.0000    | 38.2500   |           |
| 4          | 61.5833      | 29.0000        | 16.0669             | 15.2775             | 16.2464     | 14.8237   | 15.0307   | 19.5004   |
| 3.0543     | 0.0726       | 0.0190         |                     | 2.0860              |             | 0.2676    | 48.7041   |           |
| 5          | 4.2000       | 3.2000         | 12.8889             | 2.2222              | 56.8889     | 17.7778   | 0.0000    | 10.2222   |
| 0.0000     | 0.0000       | 0.0000         |                     | 1.2500              |             | 0.0000    | 41.0400   |           |
| 6          | 5.0000       | 2.0000         | 0.0000              | 0.0000              | 20.0000     | 40.0000   | 20.0000   | 0.0000    |
| 20.0000    | 0.0000       | 0.0000         |                     | 2.5000              |             | 1.0000    | 64.8000   |           |
| 7          | 1.0000       | 1.0000         | 0.0000              | 0.0000              | 0.0000      | 100.0000  | 0.0000    | 0.0000    |
| 0.0000     | 0.0000       | 0.0000         |                     | 1.0000              |             | 0.0000    | 58.3333   |           |

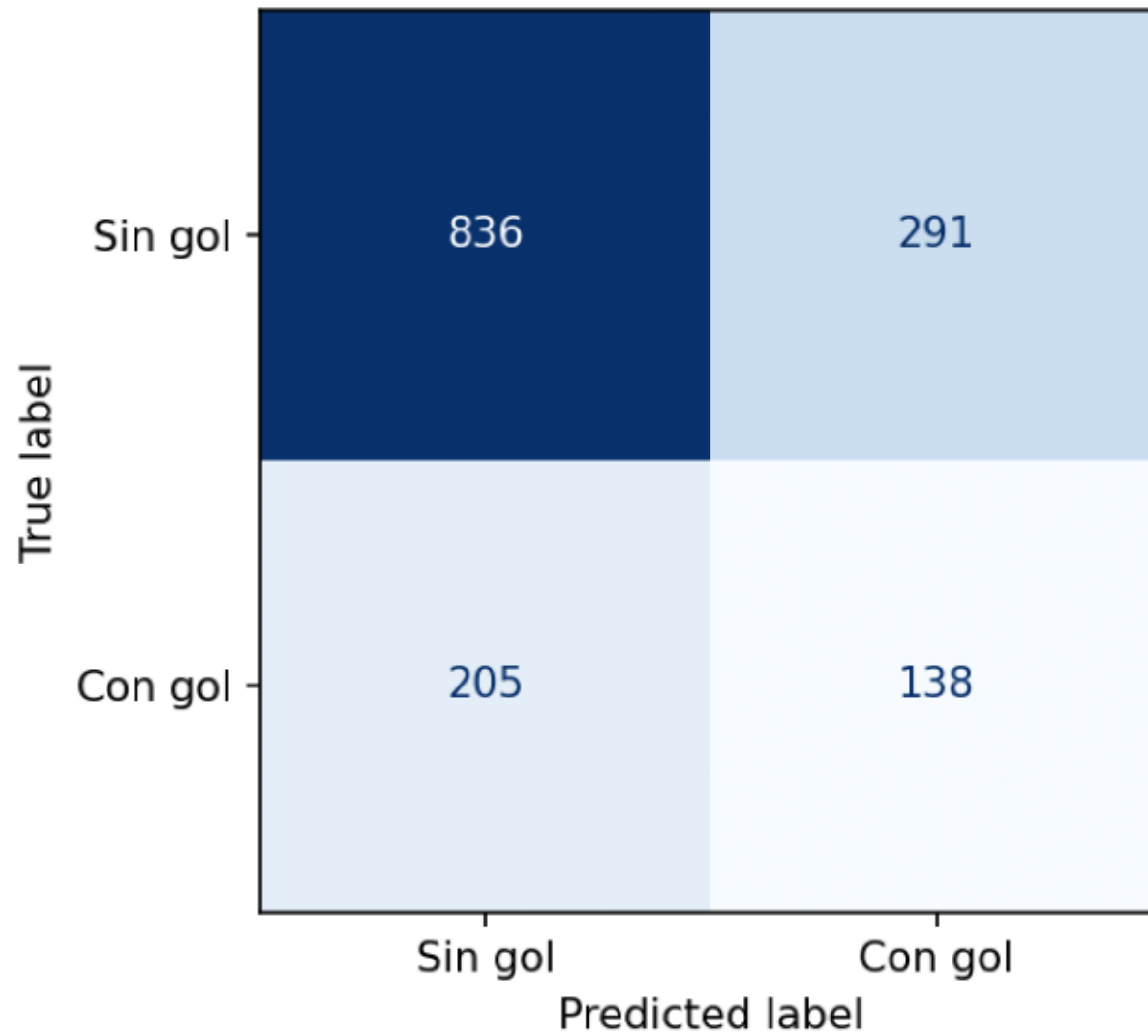
clusters\_pca.png



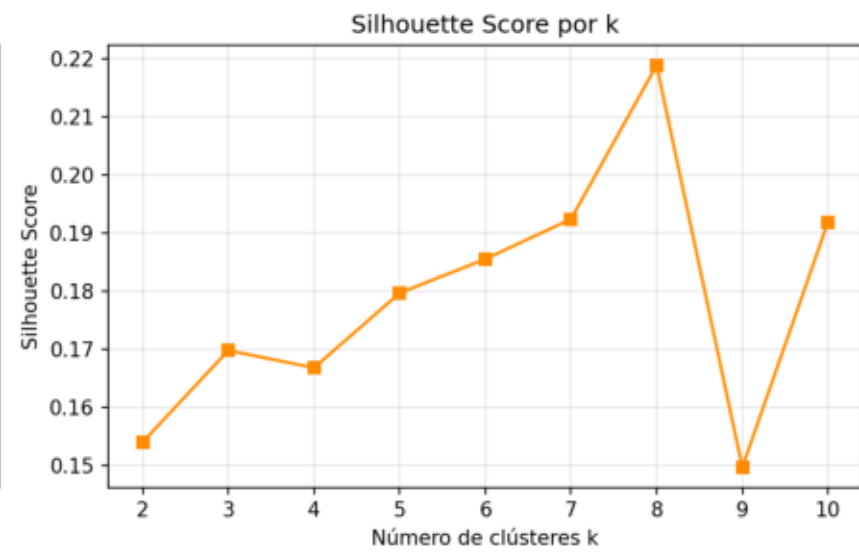
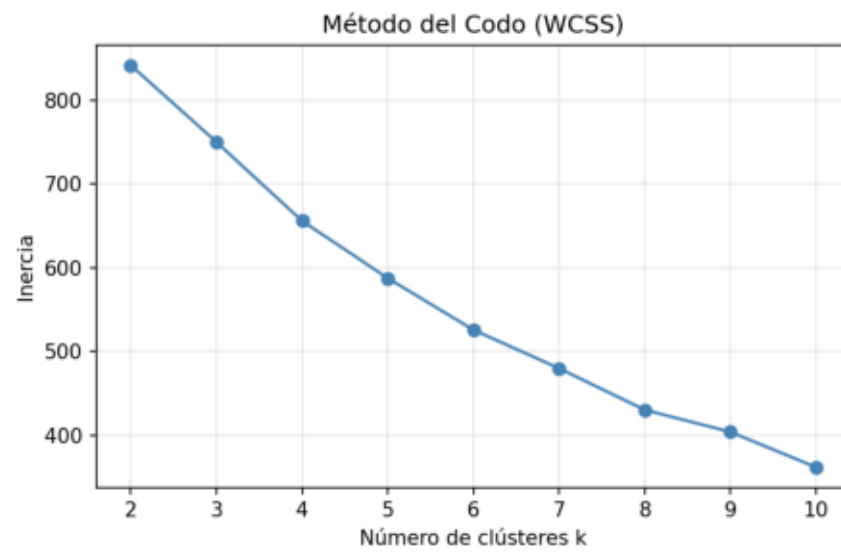
## Matriz de Confusión — Random\_Forest



## Matriz de Confusión — XGBoost

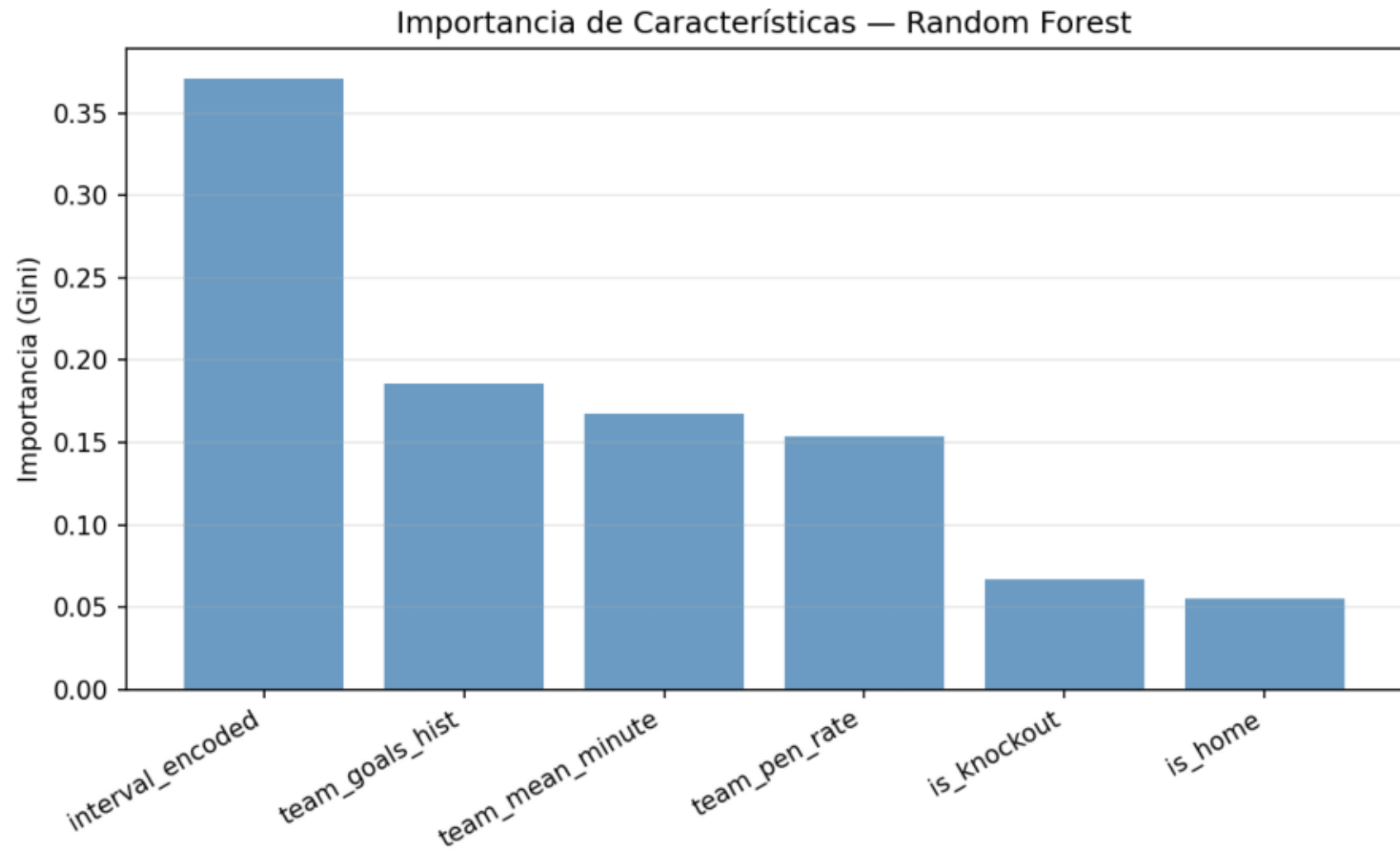


elbow\_silhouette.png



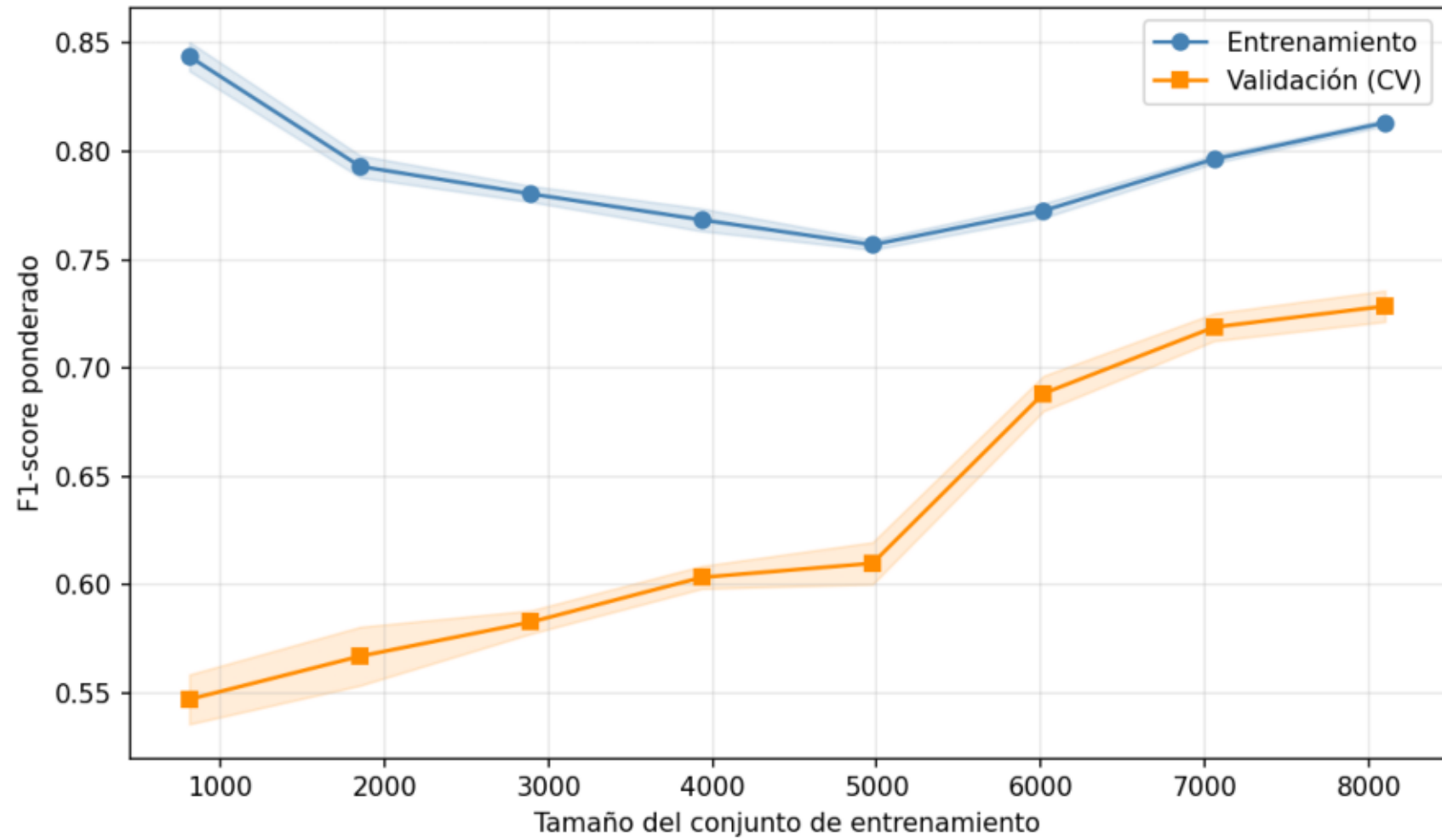


feature\_importance.png

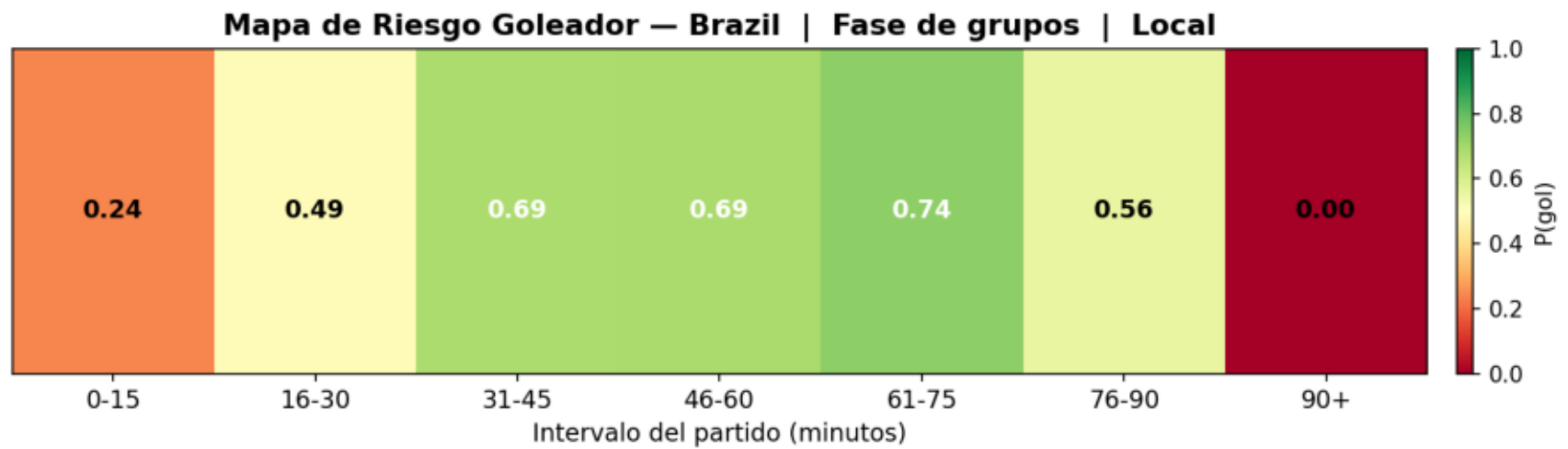


learning\_curve\_Random\_Forest.png

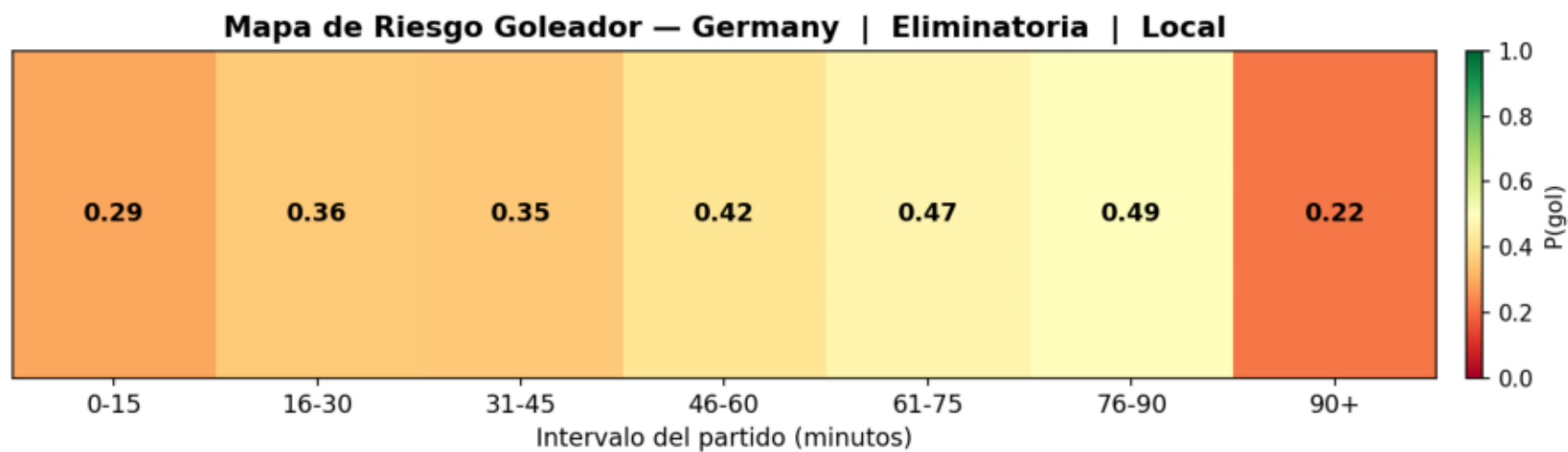
### Curvas de Aprendizaje — Random Forest



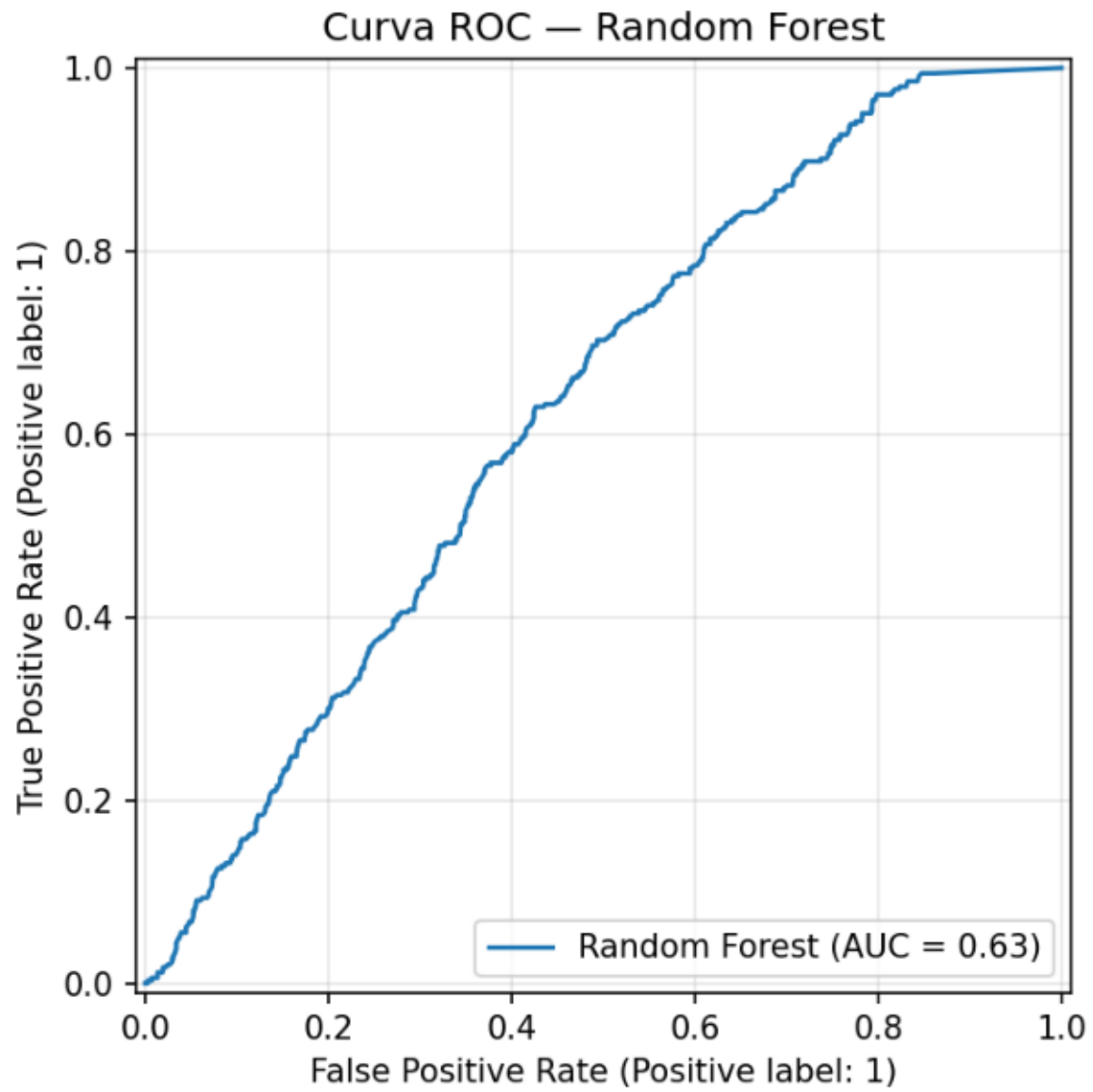
risk\_map\_Brazil.png



risk\_map\_Germany.png



roc\_curve\_Random\_Forest.png



roc\_curve\_XGBoost.png

